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Unit III: Cell biology and Biomolecules

Chapter 6

Cell: The Unit of Life

Learning Objectives

The learner will be able to,

- *Describe the cell and contributions of early scientist towards its discovery*
- *Appreciate the use of light and electron microscopes for better understanding of the cell*
- *Understand the ideas of cell theory and the different concepts associated with it*
- *Distinguish the significant characters of various groups of life forms*
- *Recognize the basic structure of cell and differentiate the cells of animals, plants, bacteria and viruses*
- *Explain the structure and functions of cell organelles including nucleus*
- *Recognize the structure of chromosome and its types*
- *Describe the flagellar structure, types and movements*
- *Get acquainted with the cytological techniques*

Chapter Outline

5QSR7L

6.1. Discovery
6.2. Microscopy
6.3. Cell theory
6.4. Cell types
6.5. Plant cell and Animal cell
6.6. Cell organelles
6.7. Nucleus
6.8. Flagella
6.9. Cytological techniques

The word '*cell*' comes from the Latin word '*Celle*" which means 'a small compartment'. The word cell was first used by Robert Hooke (1662) therefore the term '*cell*' is as old as 300 years.



6.1. Discovery

Aristotle (384-322BC), was the one who first recognised that animals and plants consists of organised structural units but unable to explain what it was. In 1660's **Robert Hooke** observed something which looks like 'honeycomb with a great little boxes' which was later called as '**cell**' from the cork tissue in 1665. He compiled his work as **Micrographia**. Later, **Antonie von Leeuwenhoek** observed unicellular particles which he named as '***animacules***'. **Robert Brown** (1831-39) described the spherical body in the plant